

Image Acquisition Method

01. Video Capture Devices

- **iPhone 12 Pro** – Triple 12MP rear cameras, 4K video at 60 FPS, excellent daylight performance.
- **iPhone XS** – Dual 12MP rear cameras, Full HD video at 30 FPS, good optical image stabilization.
- **iPhone 7 Plus** – Dual 12MP rear cameras, Full HD video at 30 FPS, strong daylight capture.
- **Samsung A05s** – 50MP main rear camera, HD video at 30 FPS, clear natural light recording.
- **Honor X5 Plus** – Dual 50MP + 2MP rear cameras, smooth HD video capture in daylight.

All devices use rear cameras for better image quality and focus, and were chosen because they capture clear video in sunlight and handle handled motion well, making them suitable for building road damage datasets.

02. Recording Parameters and Significance

- **Resolution:** iPhone 12 Pro (2160×3840), iPhone Xs and iPhone 7 Plus (1920×1080), Samsung A05s (1280×720)
High resolution captures fine details of road damages; moderate resolution reduces processing time and storage.
- **Frame Rate:** iPhone 12 Pro (60 FPS), iPhone 7 Plus & Samsung A05s (30 FPS). Higher frame rates provide smoother motion and reduce motion blur.
- **Duration:** Each video lasted **15–30 seconds**, sufficient to cover road segments while maintaining manageable file sizes

03. Environmental Conditions

Videos were recorded under afternoon natural day light, with dry roads. Surrounding features included trees, shadows, and occasional vehicles. Capturing realistic environmental conditions ensures that the dataset represents actual operational scenarios.

04. Camera Placement and Motion

Cameras were handheld with a slightly downward angle, approximately 1 meter above the road surface. Angles varied slightly between videos to capture multiple perspectives. Handheld motion enabled flexible recording, though multiple takes were required for clarity.

5. Site Selection

Videos were recorded along the **Wewalwaththa to Alupola Main Road**, a rural road. We regularly travel this road and experienced discomfort due to poor road conditions, motivating this selection to produce practical and productive results.

- **Road Type:** Rural
- **Visible Damages:** Potholes, edge cracking, missing cross line markings, traversal cracking
- **Traffic Level:** Low, minimizing interference during recording

This site ensures the dataset reflects real rural road conditions, useful for automated road inspection systems.

6. Recording Process

Cameras were held while moving slowly along the road. Multiple takes were recorded to ensure clear and usable videos. Videos affected by blurriness, shadows, or noise were discarded. Approximately 30 recordings were made, with only high-quality videos retained. We removed footage where sand obscured defects to ensure only clear, high-quality data was included.

7. Video Analysis and Dataset Preparation

- **Video Format:** MP4
- **Frame Extraction:** We will be used OpenCV to split videos into individual frames using a sampling method. We will review the frames for clarity and discard low quality ones. Selected frames will then be organized into a structured dataset, ready for model training.

This ensures a clean, consistent dataset suitable for detecting road damages.

8.Data Annotation Process

Frames were labeled for road damages, including potholes, cracks, and missing markings. The process included:

- **Object Identification:** Locating damages in each frame
- **Labeling:** Assigning appropriate damage categories
- **Quality Control:** Verifying consistency across frames

Accurate annotation is critical for training machine learning models to detect and classify road damages effectively.

9. Ethical and Safety Considerations

Recordings were conducted safely in public areas. Privacy was maintained by avoiding faces or private information. Careful camera handling ensured safe recording practices during handheld video capture.

The recorded videos and annotated dataset provide accurate information on rural road conditions. High quality capture and proper annotation ensure reliable data for machine learning models. This dataset can support the development of automated road damage detection systems.

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